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(54) **ADDITIVE COMPRISED IN CATHODE MATERIAL FOR SECONDARY BATTERY, CATHODE MATERIAL COMPRISING THE SAME, AND SECONDARY BATTERY COMPRISING CATHODE MATERIAL**

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ABSTRACT

Provided is an irreversible additive contained in a cathode material for a secondary battery, wherein the irreversible additive is an oxide represented by the following Chemical Formula 1, and wherein the oxide has a trigonal crystal structure, a cathode material including the same, and a secondary battery including the cathode material,



in Chemical Formula 1, $-0.2 \leq a \leq 0.2$, $0 \leq b < 0.5$, $0 \leq c \leq 0.2$, and

M is one or more elements selected from the group consisting of Cu, Mg, Pt, Al, Co, P, and B.

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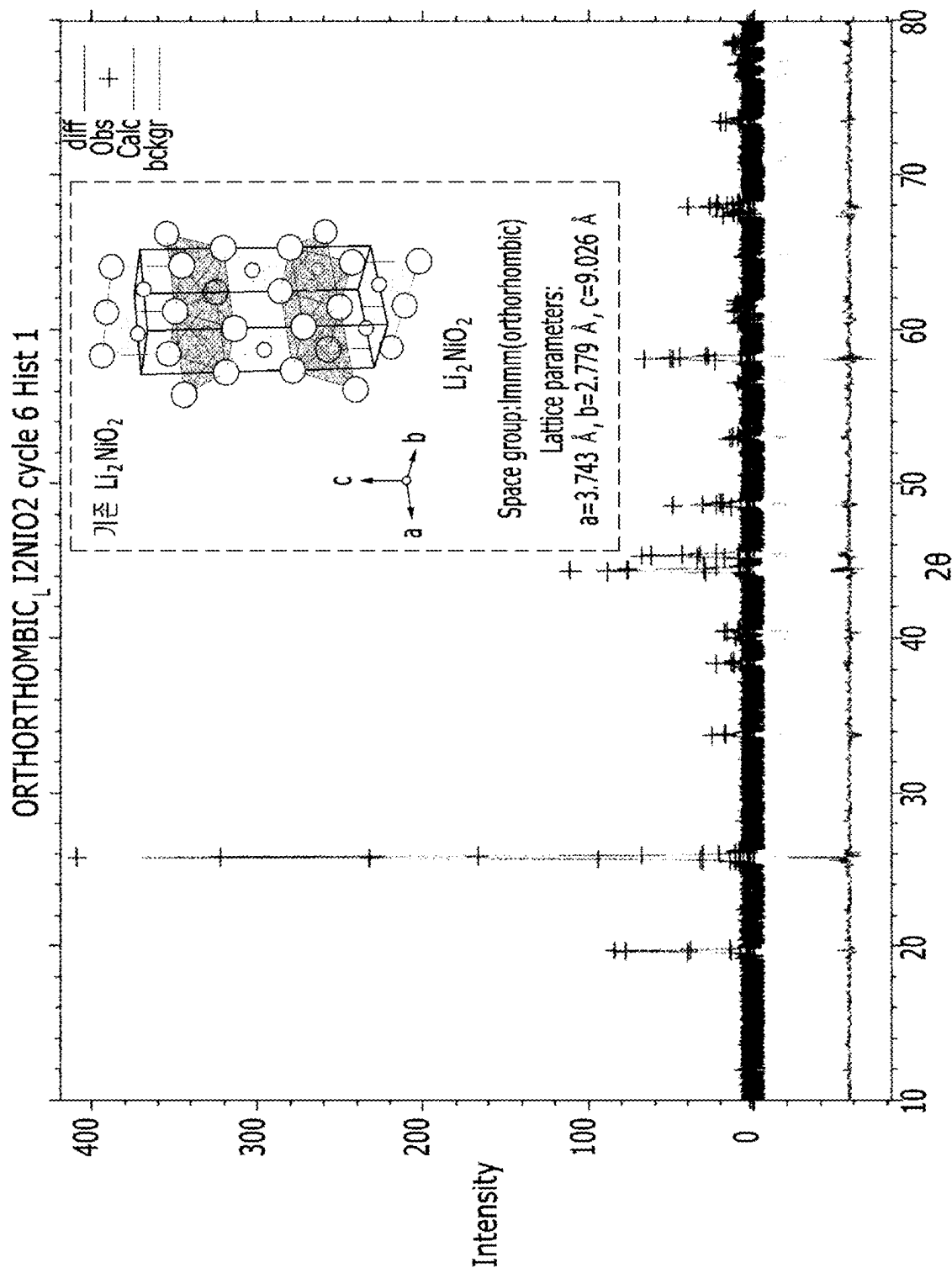
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FIG. 1



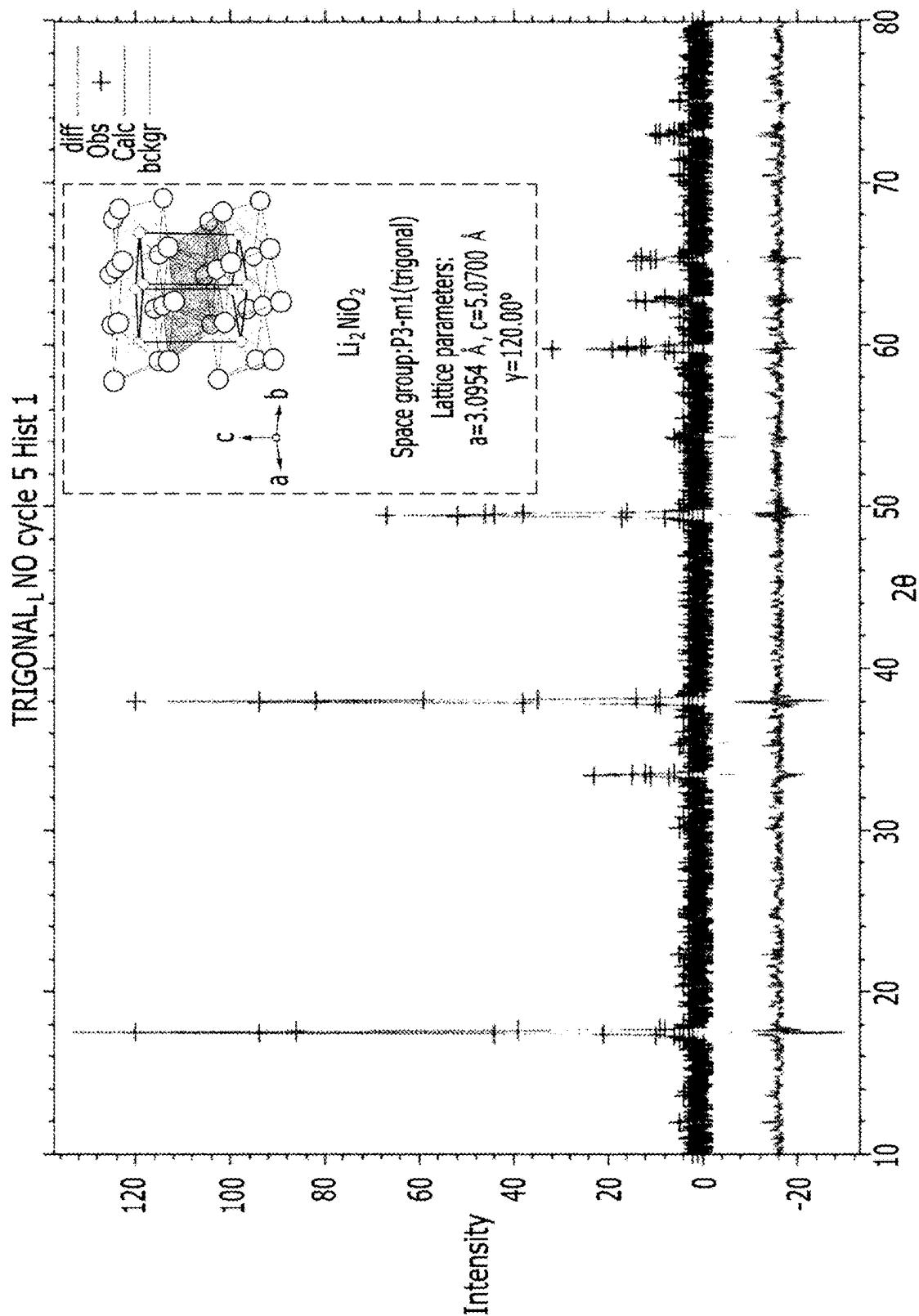


FIG. 2

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ADDITIVE COMPRISED IN CATHODE MATERIAL FOR SECONDARY BATTERY, CATHODE MATERIAL COMPRISING THE SAME, AND SECONDARY BATTERY COMPRISING CATHODE MATERIAL

CROSS REFERENCE TO RELATED APPLICATION(S)

The present application is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/KR2020/010968, filed Aug. 18, 2020, which claims priority from Korean Patent Application No. 10-2019-0116296 filed on Sep. 20, 2019, the disclosures of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to an irreversible additive contained in a cathode material for a secondary battery that can minimize generation of impurities or gas, a cathode material including the same, and a secondary battery including the cathode material.

BACKGROUND ART

Due to the rapid increase in the use of fossil fuels, the demand for the use of alternative energy or clean energy is increasing, and as part thereof, the fields that are being studied most actively are the fields of power generation and power storage using electrochemistry.

At present, a secondary battery is a representative example of an electrochemical device that utilizes such electrochemical energy, and the range of use thereof tends to be gradually expanding.

Recently, with the increase of the technological development and demand for mobile devices such as portable computers, portable phones, and cameras, demand for secondary batteries as an energy source rapidly increases. Among such secondary batteries, many studies have been conducted on a lithium secondary battery that exhibits high energy density and operating voltage, a long cycle life, and a low self-discharge rate, and has been commercialized and widely used.

In addition, as interest in environmental issues grows, studies are frequently conducted on an electric vehicle, a hybrid electric vehicle, etc. which can replace a vehicle using fossil fuels such as a gasoline vehicle and a diesel vehicle, which are one of the main causes of air pollution. Although a nickel metal hydride secondary battery is mainly used as a power source for the electric vehicle and the hybrid electric vehicle, research on the use of a lithium secondary battery having high energy density and discharge voltage is actively being conducted, a part of which are in the commercialization stage.

Carbon materials are mainly used as an anode active material of such lithium secondary battery, and lithium transition metal composite oxide is used as a cathode active material of lithium the secondary battery. Among them, in addition to lithium cobalt composite metal oxides such as LiCoO_2 having high operating voltage and excellent capacity characteristics, various lithium transition metal oxides such as LiNiO_2 , LiMnO_2 , LiMn_2O_4 or LiFePO_4 have been developed.

Meanwhile, due to the consumption of Li ions during the initial sage of charging and discharging, the formation of a

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solid electrolyte interphase (SEI) layer and the irreversibility of cathode and anode ions can occur.

Consequently, the energy density is reduced, and there is a problem that the theoretical amount that can be designed cannot be sufficiently used.

In order to solve these problems, an irreversible additive was added to the cathode material to supplement lithium ions. However, Li_2NiO_2 , which is a commonly used irreversible additive, had an orthorhombic crystal structure and belonged to a space group of Immm. However, the above material has a problem of causing the generation of impurities or gas while undergoing three stages of structural changes in the operating voltage range after the initial charge of the secondary battery.

Specifically, the above material maintains an orthorhombic crystal structure in the range of 3.0 to 3.5V, but depending on the desorption of Li, the material undergoes three-stage crystal structure changes to a trigonal system at 3.5 to 4.0 V and to a monoclinic system at 3.5 to 4.25 V. When such material is decomposed, there is a problem that impurities or gas may be generated.

In particular, when the irreversible additive (Li_2NiO_2) having an orthorhombic crystal structure leads to unpredictable by-products and extra gas generation when the crystal structure changes to a trigonal system.

Therefore, there is an urgent need for a technique that solves the above problems and does not cause generation of impurities or gas within the operating voltage range of the secondary battery, even while sufficiently expressing Li ions at the initial stage of charging.

DETAILED DESCRIPTION OF THE PRESENT DISCLOSURE

Technical Problem

The present disclosure has been designed to solve the above-mentioned problems, and an object of the present disclosure is to provide an irreversible additive that can minimize generation of impurities or gas in the working voltage range of a secondary battery, even while providing Li ions at the initial stage of charging to reduce irreversibility.

Another object of the present disclosure is to provide a cathode material for a secondary battery containing the irreversible additive, and a secondary battery exhibiting excellent electrochemical properties by including the same.

Technical Solution

It should be understood that the terms and wordings used herein should not be construed to be limited to general or lexical means, nor should the concepts of the terms be considered to be defining or describing the present disclosure made by the inventor(s) in the best way. Further, the terms and wordings should be constructed to have meanings and concepts that agree with the technical spirit of the present disclosure.

Through the following disclosure and examples, certain embodiments of an irreversible additive contained in a cathode material for a secondary battery, a cathode material including the irreversible additive, and a secondary battery including the cathode material, as well as a secondary battery including the same will be described. These particular described embodiments and examples are for illustration only and the present disclosure should not be considered to be limited to such embodiments and examples herein.

The irreversible additive contained in a cathode material for a secondary battery according to one embodiment of the present disclosure is an oxide represented by the following Chemical Formula 1, wherein the oxide has a trigonal crystal structure:



in Chemical Formula 1, $-0.2 \leq a \leq 0.2$, $0 \leq b < 0.5$, $0 \leq c \leq 0.2$, and

M is one or more elements selected from the group consisting of Cu, Mg, Pt, Al, Co, P, and B.

In this regard, conventionally, the oxide having Chemical Formula 1 were prepared by mixing a lithium raw material, a nickel raw material, and a raw material of element M, and then heat-treating them.

When the general raw materials were mixed and heat-treated in this way, the oxide was produced into a material having an orthorhombic crystal structure, which is the most stable form. Therefore, conventionally, an oxide having an orthorhombic crystal structure has been added as the irreversible additive.

However, as a result of repeated in-depth studies, the present inventors have found that the material having the crystal structure as described above is added, impurities or gas are generated within the working voltage range of the secondary battery, and thus there is a problem that the battery performance is rapidly deteriorated as the cycle proceeds, confirmed that the above causes occur as a result of undergoing three stages of crystal structure changes as mentioned above, and found out that when the oxide represented by Chemical Formula 1 is added as a material having a trigonal crystal structure from the beginning, the trigonal and monoclinic crystal structures can be reversibly maintained according to the voltage of the secondary battery within the operating voltage range of the secondary battery.

On the other hand, when the structure changes according to the voltage in the secondary battery, a trigonal crystal structure causes desorption of Li ions. When the molar ratio of Li ions is 1, it produces LiNiO_2 and has the crystal structure.

However, according to the present disclosure, the irreversible additive is an excess lithiated oxide in which the molar ratio of Li is about twice the transition metal, as expressed by Formula 1 above. More specifically, the irreversible additive may be Li_2NiO_2 .

Further, particularly, the irreversible additive may belong to a space group of P3-m1, and more particularly, the crystal lattice of the oxide may be $a=3.0954 \text{ \AA}$, $c=5.0700 \text{ \AA}$, $\gamma=120.00^\circ$.

In one embodiment, the material as above is prepared by a process in which LiNiO_2 is mixed with Li+benzophenone- and reacted under THF to obtain a trigonal Li_2NiO_2 having weak crystallinity, which is then heat-treated under an inert atmosphere to obtain a trigonal Li_2NiO_2 having high crystallinity.

The reaction under THF is specifically performed by a process in which the mixture is stirred, filtered, washed with dry THF, and then dried under vacuum.

The heat treatment is performed at 200 to 400° C. for 10 to 24 hours under an inert atmosphere.

The inert atmosphere may be a helium or argon atmosphere, and the heat treatment is performed while flowing the gases.

Further, continuing with this exemplary embodiment, the preparation should be performed within the temperature and time range during the heat treatment, so that only crystallinity can be improved without changing the crystal structure

of the trigonal Li_2NiO_2 formed by reacting under THF. When the temperature is too low or the time is short, the crystallinity is not sufficiently improved, and when the temperature is too high or the time is long, the crystal structure itself may be changed, which is not preferable.

The LiNiO_2 can be produced by a conventionally known production method.

For example, it is produced by mixing a lithium raw material and a nickel raw material in a molar ratio satisfying the composition ratio and heat-treating them.

The heat treatment is performed at 650 to 800° C. for 12 to 36 hours under an air atmosphere. In the case of wet method, a drying process may be further included.

The preparation must be performed within the temperature and time of the heat treatment, so that the reaction between the lithium raw material and the nickel raw material can sufficiently occur, and unreacted materials can be minimized.

As to the lithium raw material, lithium-containing oxides, sulfates, nitrates, acetates, carbonates, oxalates, citrates, halides, hydroxides or oxyhydroxides, and the like can be used, and specific examples thereof include Li_2CO_3 , LiNO_3 , LiNO_2 , LiOH , $\text{LiOH} \cdot \text{H}_2\text{O}$, LiH , LiF , LiCl , LiBr , LiI , CH_3COOLi , Li_2O , Li_2SO_4 , CH_3COOLi , or $\text{Li}_3\text{C}_6\text{H}_5\text{O}_7$. Any one or a mixture of two or more of them may be used.

As the nickel raw material, nickel-containing oxides, sulfates, nitrates, acetates, carbonates, oxalates, citrates, halides, hydroxides or oxyhydroxides, and the like can be used, and specific examples thereof include NiO , $\text{Ni}(\text{NO}_3)_2$, LiNiO_2 , NiSO_4 , $\text{Ni}(\text{OH})_2$, and the like. Any one or a mixture of two or more of them may be used.

The irreversible additive of the crystal structure as above provides sufficient Li at the initial stage of charging with an excessive amount of lithium, thereby being able to solve the irreversibility problem, and also reduce the stage of change in crystal structure within the operating voltage range, thereby being able to minimize incidental problems such as generation of impurities or gas resulting from desorption of an excessive amount of Li ions.

Meanwhile, according to one embodiment of the present disclosure, there is provided a cathode material including the irreversible additive and a cathode active material.

Continuing with this embodiment, the content of the irreversible additive may be 0.1% by weight to 10% by weight, more specifically 1% by weight to 5% by weight, and more specifically 1% to 3% by weight based on the total weight of the cathode material.

When the content of the irreversible additive is less than 0.1% by weight outside the above range, the compensation effect of anode efficiency due to the addition of an irreversible additive cannot be obtained, and when the content exceeds 10% by weight, problems such as volume expansion of the electrode due to the generation of impurities or gas, and deterioration of life may occur.

Further, according to one embodiment of the present disclosure, there is provided a secondary battery comprising cathode in which a cathode material is coated onto a cathode current collector,

wherein the cathode material includes an irreversible additive including an oxide represented by the following Chemical Formula 1, and a cathode active material, and

wherein the irreversible additive has a trigonal crystal system and is converted into a monoclinic crystal system in a reversible structural conversion manner within a range in which the operating range of the secondary battery is 4.0V or more.

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in Chemical Formula 1, $-0.2 \leq a \leq 0.2$, $0 \leq b < 0.5$, $0 \leq c \leq 0.2$, and

M is one or more elements selected from the group consisting of Cu, Mg, Pt, Al, Co, P, and B.

As described above, the crystal structure of the irreversible additive of lithium nickel oxide changes in the operating voltage range of the secondary battery, and this is also the same as when the irreversible additive, depending on the present disclosure is used.

Therefore, according to the present disclosure, even if the oxide represented by Chemical Formula 1 having a trigonal crystal structure is added as an irreversible additive, the crystal structure of the oxide can change to a monoclinic crystal system within the operating voltage range of the secondary battery according to the insertion and desorption of Li ions.

In other words, the irreversible additive according to the present disclosure is added to the cathode material in the form of a trigonal crystal structure, and can be reversibly converted into a monoclinic crystal system within the operating voltage range of the secondary battery.

In this case, the oxide having a monoclinic crystal structure may, specifically, belong to a space group of $C_{2/m}$.

On the other hand, the cathode active material contained in the cathode material may be, for example, LiCoO_2 , LiNiO_2 , LiMnO_2 , LiMn_2O_2 , $\text{Li}(\text{Ni}_a\text{Co}_b\text{Mn}_c)\text{O}_2$ ($0 < a < 1$, $0 < b < 1$, $0 < c < 1$, $a+b+c=1$), $\text{LiNi}_{1-d}\text{Co}_d\text{O}_2$, $\text{LiCo}_{1-d}\text{Mn}_d\text{O}_2$, $\text{LiNi}_{1-d}\text{Mn}_d\text{O}_2$ ($0 \leq d < 1$), $\text{Li}(\text{Ni}_a\text{Co}_b\text{Mn}_c)\text{O}_4$ ($0 < a < 2$, $0 < b < 2$, $0 < c < 2$, $a+b+c=2$), $\text{LiMn}_{2-e}\text{Ni}_e\text{O}_4$, $\text{LiMn}_{2-e}\text{Co}_e\text{O}_4$ ($0 < e < 2$), LiCoPO_4 , LiFePO_4 , or the like, and any one or a mixture of two or more of them may be used.

Of these, in one particular example, the cathode active material may include an oxide represented by the following Chemical Formula 2:



in Chemical Formula 2, $0 < a < 1$, $0 < b < 1$, $0 < c < 1$, $a+b+c=1$.

In the oxide of Chemical Formula 2, the crystal structure easily changes from hexagonal to monoclinic while Li ions are desorbed and inserted in the operating voltage range of the secondary battery. Therefore, since the oxide can have a structure similar to that of the irreversible additive of the present disclosure within the operating range, it is more effective in the use of the irreversible additive according to the present disclosure.

More specifically, the oxide represented by Chemical Formula 2 may be contained in an amount of 80% by weight or more based on the total weight of the cathode active material.

The cathode material may further include a conductive material, a binder, a filler and the like, in addition to the cathode active material and the irreversible additive.

The conductive material is used to impart conductivity to the electrode, and in the battery to be configured, the conductive material can be used without particular limitation as long as it does not cause chemical changes and has electronic conductivity. Particular examples include carbon-based materials such as carbon black, acetylene black, ketjen black, channel black, furnace black, lamp black, thermal black and carbon fiber; graphite such as natural graphite and artificial graphite; metal powder or metal fibers such as copper, nickel, aluminum and silver; conductive whiskey such as zinc oxide and potassium titanate; conductive metal oxides such as titanium oxide; or a conductive polymer such as a polyphenylene derivative. Any one alone or a mixture of two or more of them may be used. The conductive

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material may be included in an amount of 1% to 30% by weight based on the total weight of the cathode material.

The binder plays a role of improving adhesion between the cathode active material particles and adhesive strength between the cathode active material and the current collector. Specific examples include polyvinylidene fluoride (PVDF), vinylidene fluoride-hexafluoropropylene copolymer (PVDF-co-HFP), polyvinyl alcohol, polyacrylonitrile, carboxymethylcellulose (CMC), starch, hydroxypropylcellulose, recycled cellulose, polyvinylpyrrolidone, tetrafluoroethylene, polyethylene, polypropylene, ethylene-propylene-diene polymer (EPDM), sulfonated-EPDM, styrene butadiene rubber (SBR), fluororubber, or various copolymers thereof, and any one alone or a mixture of two or more of them may be used. The binder may be included in an amount of 1% to 30% by weight based on the total weight of the cathode material.

The cathode current collector is not particularly limited as long as it has conductivity while not causing chemical changes to the battery, and for example, stainless steel, aluminum, nickel, titanium, calcined carbon, or aluminum or stainless steel having a surface treated with carbon, nickel, titanium, silver, etc. may be used. In addition, the cathode current collector may have a thickness of 3 μm to 500 μm , and may have fine irregularities formed on the surface thereof to increase the adhesion of the cathode active material. For example, it may be used in various forms such as films, sheets, foils, nets, porous bodies, foams, and nonwoven fabrics.

The secondary battery may have a structure in which an electrode assembly is built in a battery case together with an electrolyte, with the electrode assembly including:

the cathode;

an anode in which an anode material including an anode active material is coated onto an anode current collector; and

a separator that is interposed between the cathode and the anode.

Specifically, the secondary battery may be a lithium secondary battery.

The anode may also be manufactured in a form in which an anode material including an anode active material is applied on an anode current collector, and the anode material may further include a conductive material and a binder as described above, together with an anode active material.

As to the anode active material, a compound capable of reversibly intercalating and deintercalating lithium may be used. Specific examples thereof may include carbonaceous materials such as artificial graphite, natural graphite, graphitized carbon fibers and amorphous carbon; metallic substances capable of alloying with lithium, such as Si, Al, Sn, Pb, Zn, Bi, In, Mg, Ga, Cd, Si alloy, Sn alloy or Al alloy; metal oxides capable of doping and undoping lithium, such as SiO_x ($0 < x < 2$), SnO_2 , vanadium oxide and lithium vanadium oxide; or a composite including the above metallic substances and the carbonaceous material such as a Si—C composite or a Sn—C composite, or the like, and any one or a mixture of two or more of them may be used. In addition, a metal lithium thin film may be used as the anode active material. Further, both low crystalline carbon and high crystalline carbon may be used as the carbon material. Typical examples of the low crystalline carbon may be soft carbon and hard carbon. Typical examples of the high crystalline carbon may be amorphous, planar, flaky, spherical or fibrous natural or artificial graphite, Kish graphite, pyrolytic carbon, mesophase pitch based carbon fiber, meso-

carbon microbeads, mesophase pitches, and high-temperature calcined carbon such as petroleum or coal tar pitch derived cokes.

The anode current collector is not particularly limited as long as it has high conductivity without causing chemical changes to the battery. For example, copper, stainless steel, aluminum, nickel, titanium, calcined carbon, copper or stainless steel having a surface treated with carbon, nickel, titanium, silver, etc., aluminum-cadmium alloy, and the like may be used. In addition, the anode current collector may generally have a thickness of 3 μm to 500 μm , and, like the cathode current collector, may have fine irregularities formed on the surface thereof to enhance the bonding strength of the anode active material. For example, it may be used in various forms such as films, sheets, foils, nets, porous bodies, foams and nonwoven fabrics.

The separator separates the anode and the cathode, and provides a passage for lithium ions to move. Any separator may be used without particular limitation as long as it is generally used as a separator in a lithium secondary battery. Particularly, a separator having excellent moisture-retention ability for an electrolyte while having low resistance to the migration of electrolyte ions is preferable. Specifically, a porous polymer film, for example, a porous polymer film made of polyolefin-based polymers such as ethylene homopolymer, propylene homopolymer, ethylene/butene copolymer, ethylene/hexene copolymer, and ethylene/methacrylate copolymer, or a laminated structure having two or more layers thereof may be used. In addition, a conventional porous nonwoven fabric, for example, a nonwoven fabric made of high melting point glass fiber, polyethylene terephthalate fiber, or the like may also be used. In addition, in order to secure heat resistance or mechanical strength, a coated separator containing a ceramic component or a polymer material may be used, and optionally, a single layer or a multilayer structure may be used.

In addition, the electrolyte used in the present disclosure may include, but is not limited to, an organic liquid electrolyte, an inorganic liquid electrolyte, a solid polymer electrolyte, a gel polymer electrolyte, a solid inorganic electrolyte, a molten inorganic electrolyte or the like which can be used in the preparation of a lithium secondary battery.

Specifically, the electrolyte may include an organic solvent and a lithium salt.

As to the organic solvent, any solvent may be used without particular limitation as long as it can serve as a medium through which ions involved in the electrochemical reaction of the battery can move. Specifically, as the organic solvent, an ester-based solvent such as methyl acetate, ethyl acetate, γ -butyrolactone, or ϵ -caprolactone; an ether-based solvent such as dibutyl ether or tetrahydrofuran; a ketone-based solvent such as cyclohexanone; an aromatic hydrocarbon-based solvent such as benzene or fluorobenzene; a carbonate-based solvent such as dimethyl carbonate (DMC), diethylcarbonate (DEC), methylethylcarbonate (MEC), ethylmethylcarbonate (EMC), ethylene carbonate (EC), or propylene carbonate (PC); an alcohol-based solvent such as ethyl alcohol or isopropyl alcohol; nitriles such as R-CN (R is a straight, branched or cyclic C2-C20 hydrocarbon group, and may include a double-bond aromatic ring or ether bond); amides such as dimethylformamide; dioxolanes such as 1,3-dioxolane; or sulfolanes may be used. Among them, the carbonate-based solvent is preferable, and a mixture of a cyclic carbonate (e.g., ethylene carbonate, propylene carbonate, etc.) having high ionic conductivity and a high-dielectric constant, which may increase charge/discharge performance of the battery, and a low-viscosity linear car-

bonate-based compound (e.g., ethyl methyl carbonate, dimethyl carbonate, diethyl carbonate, etc.) may be more preferably used. In this case, when the cyclic carbonate and the chain carbonate are mixed and used in a volume ratio of about 1:1 to about 1:9, the electrolyte may exhibit excellent performance.

Various lithium salts may be used without particular limitation as long as it is a compound capable of providing lithium ions used in a lithium secondary battery. Specifically, LiPF_6 , LiClO_4 , LiAsF_6 , LiBF_4 , LiSbF_6 , LiAlO_4 , LiAlCl_4 , LiCF_3SO_3 , $\text{LiC}_4\text{F}_9\text{SO}_3$, $\text{LiN}(\text{C}_2\text{F}_5\text{SO}_3)_2$, $\text{LiN}(\text{C}_2\text{F}_5\text{SO}_2)_2$, $\text{LiN}(\text{CF}_3\text{SO}_2)_2$, LiCl , LiI , $\text{LiB}(\text{C}_2\text{O}_4)_2$, or the like may be used as the lithium salt. It is preferable to use the lithium salt in a concentration range of 0.1 to 2.0 M. If the concentration of the lithium salt is within the above range, since the electrolyte has an appropriate conductivity and viscosity, excellent electrolyte performance can be exhibited, and lithium ions can effectively move.

In order to improve the lifespan characteristics of the battery, suppress a reduction in battery capacity and improve discharge capacity of the battery, for example, one or more additives such as halo-alkylene carbonate-based compound such as difluoroethylene carbonate, pyridine, triethylphosphite, triethanolamine, cyclic ether, ethylene diamine, n-glyme, hexaphosphoric triamide, a nitrobenzene derivative, sulfur, a quinone imine dye, N-substituted oxazolidinones, N,N-substituted imidazolidine, ethylene glycol dialkyl ether, an ammonium salt, pyrrole, 2-methoxy ethanol, or aluminum trichloride may be further added to the electrolyte in addition to the above electrolyte components. In this case, the additive may be included in an amount of 0.1% to 5% by weight based on the total weight of the electrolyte.

The lithium secondary battery according to the present disclosure as described above may be used as a power source of devices in portable devices such as mobile phones, notebook computers, digital cameras, and electric vehicles such as hybrid electric vehicles (HEVs).

Advantageous Effects

In one particular exemplary embodiment, the irreversible additive according to the present disclosure is the oxide represented by Chemical Formula 1, and has a trigonal crystal structure, so that within the operating voltage range of the secondary battery, the problem of generation of impurities or gas due to the desorption of excess Li ions can be significantly reduced. Accordingly, a lithium secondary battery manufactured by using a cathode material including the same can effectively compensate for irreversibility and also exhibit increased electrochemical characteristics and lifespan characteristics.

DETAILED DESCRIPTION OF THE EMBODIMENTS

FIG. 1 is a graph showing the XRD measurement results of Comparative Example 1 according to Experimental Example 1;

FIG. 2 is a graph showing the XRD measurement results of Example 1 according to Experimental Example 1.

DETAILED DESCRIPTION OF THE EMBODIMENTS

Various embodiments of the present disclosure will be described in detail with reference to the accompanying figures so that those skilled in the art can easily implement

them. The present disclosure may be modified in various ways, and is not limited to the embodiments set forth herein.

Comparative Example 1

22.9 g of Li_2O and 30 g of NiO (molar ratio 1:1) were mixed, and then heat-treated at 685 degrees Celsius for 18 hours under an N_2 atmosphere, and then the resulting reaction product was cooled to obtain irreversible additive particles Li_2NiO_2 .

Example 1

LiNiO_2 and more than 1.5M Li +benzophenone—were reacted in the presence of THF (tetrahydrofuran) under an inert atmosphere.

Specifically, the mixture of the above materials was stirred for one day, and the mixed powder was filtered. The obtained mixed powder was washed with dry THF and dried under vacuum to obtain a pre-powder in which a small amount of trigonal Li_2NiO_2 and LiNiO_2 were mixed.

Subsequently, the pre-powder was heat-treated at 225° C. for 14 hours under dry helium flow to obtain a Li_2NiO_2 powder having a trigonal crystal structure with improved crystallinity.

Experimental Example 1

2 g of the irreversible additive particles prepared in Comparative Example 1 and Example 1 were each collected as samples, and subjected to XRD analysis. The results are shown in FIGS. 1 and 2.

XRD analysis was measured with a Bruker XRD D4 instrument, and experiment was performed from 10 degrees to 80 degrees in 0.02 steps using a Cu source target.

Referring to FIGS. 1 and 2, it can be seen that Li_2NiO_2 having a different structure was formed according to Comparative Example 1 and Example 1.

Specifically, it can be seen that Comparative Example 1 is formed with an orthorhombic structure, and Example 1 is formed with a trigonal crystal structure.

Comparative Example 2 and Example 2

Using the irreversible additives prepared in Comparative Examples 1 and Example 1, a cathode and a lithium secondary battery were manufactured by the following method.

Specifically, the irreversible additive prepared in Comparative Example 1 and Example 1, $\text{LiNi}_{0.4}\text{Mn}_{0.3}\text{Co}_{0.3}\text{O}_2$ as a cathode active material, a carbon black conductive material and a PVdF binder were mixed in a weight ratio of 4.6:87.9:3.5:4 in an N-methylpyrrolidone solvent to prepare a cathode slurry. The slurry was coated onto an aluminum current collector, and dried and rolled to prepare a cathode.

In addition, MCMB (mesocarbon microbead), which is an artificial graphite mixed with 10 wt. % of SiO as an anode active material, a carbon black conductive material and PVdF binder were mixed in a weight ratio of 90:5:5 in an N-methylpyrrolidone solvent to prepare a composition for forming an anode, which was coated onto a copper current collector to prepare an anode.

A porous polyethylene separator was interposed between the cathode and the anode prepared as described above to manufacture an electrode assembly. The electrode assembly was placed inside the case, and then an electrolyte was injected into a case to manufacture a lithium secondary battery. At this time, the electrolyte was prepared by dis-

solving 1.15M lithium hexafluorophosphate (LiPF_6) in an organic solvent consisting of ethylene carbonate/dimethyl carbonate/ethylmethyl carbonate (EC/DMC/EMC mixing volume ratio=3/4/3).

Experimental Example 2

2 g of the irreversible additive particles prepared in Example 1 and Comparative Example 1 were each collected as samples, and their oxygen formation energies were measured. The results are shown in Table 1 below.

Specifically, the calculation of oxygen formation energy was performed based on the calculated value for DFT (density functional theory), PBE functional PAW_PBE pseudopotential Hubbard U term for Ni =6.2 eV, cut-off energy=520 eV, calculation model: supercell with $\text{Li}_{48}\text{Ni}_{24}\text{O}_{48}$ atoms, oxygen vacancy (VO) production concentration=1/48 (~2.1 at. %) O_2 gas (O-rich environment).

TABLE 1

Oxygen (V_O) formation energy (eV)	
Comparative Example 1	4.56
Example 1	5.63

Referring to Table 1, considering that the energy of the trigonal crystal irreversible additive of Example 1 is higher than that of the orthorhombic system of Comparative Example 1, it is presumed that the structure change in the intercalation of Li ions during charging and discharging is more robust than the orthorhombic system. Therefore, when used for a cathode as in Example 2 and Comparative Example 2, it is expected that the trigonal crystal irreversible additive will cause less side reactions than orthorhombic irreversible additives.

The invention claimed is:

1. A secondary battery comprising a cathode in which a cathode material is coated onto a cathode current collector, wherein the cathode material includes an additive including an oxide represented by the following Chemical Formula 1, and a cathode active material, wherein a content of the additive is 0.1% by weight to 10% by weight based on a total weight of the cathode material; wherein the additive initially has a trigonal crystal system and is converted into a monoclinic crystal system in a reversible structural conversion manner within a range in which the operating range of the secondary battery is 4.0V or more, wherein the additive is reversibly maintained between the trigonal crystal structures and the monoclinic crystal structures, and



in Chemical Formula 1, $-0.2 \leq a \leq 0.2$, $0 \leq b < 0.5$, $0 \leq c \leq 0.2$, and

M is one or more elements selected from the group consisting of Cu, Mg, Pt, Al, Co, P, and B.

2. The secondary battery according to claim 1, wherein the additive belongs to a space group of C2/m when the crystal structure is a monoclinic crystal system.

3. The secondary battery according to claim 1, wherein the cathode active material comprises an oxide represented by the following Chemical Formula 2,



in Chemical Formula 2, $0 < a < 1$, $0 < b < 1$, $0 < c < 1$, $a+b+c=1$.

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4. The secondary battery according to claim 1, wherein the secondary battery has a structure in which an electrode assembly is built in a battery case together with an electrolyte, with the electrode assembly including:

the cathode;

an anode in which an anode material including an anode active material is coated onto an anode current collector; and

a separator that is interposed between the cathode and the anode.

5. The secondary battery according to claim 1, wherein the secondary battery is a lithium secondary battery.

6. The secondary battery according to claim 1, wherein the oxide belongs to a space group of P3-m1.

7. The secondary battery according to claim 1, wherein the oxide has a crystal lattice of $a=3.0954 \text{ \AA}$, $c=5.0700 \text{ \AA}$, $\gamma=120.00^\circ$.

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8. The secondary battery according to claim 1, wherein the oxide is Li_2NiO_2 .

9. The secondary battery according to claim 3, wherein the cathode active material comprises nickel, cobalt, and manganese.

10. The secondary battery according to claim 3, wherein the cathode active material comprises the oxide represented by Chemical Formula 2 in an amount of 80% by weight or more based on a total weight of the cathode active material.

11. The secondary battery according to claim 1, wherein the cathode material further comprises:

a conductive material in an amount of 1% to 30% by weight based on the total weight of the cathode material; and

15 a binder in an amount of 1% to 30% by weight based on the total weight of the cathode material.

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